

Article

CNF/p(AM-co-KAA) Semi-Interpenetrating Network Hydrogel Fertilizer Carriers for Enhanced Nutrient Use Efficiency, Water Retention, and Salt–Alkali Resistance

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Abstract

Developing functional agricultural materials that synchronize nutrient release, water retention, and soil amendment is crucial to advancing resource-efficient, sustainable farming systems. However, integrating these multifunctional properties within a single material remains a significant challenge. In this work, we fabricated a multifunctional hydrogel (CPAUH) via a one-pot synthesis strategy, which was composed of carboxylated cellulose nanofibers as a rigid network combined with poly(AA-co-KAA), forming a semi-interpenetrating network (semi-IPN) for loading urea and humic acid. The structure and properties of hydrogels were characterized by FTIR, TGA, SEM, and XPS. The CPAUH exhibited outstanding mechanical strength (0.169 MPa), water absorption capacity (121.65 g g^{−1}), and retained 118 g g^{−1} after three absorption–desorption cycles, demonstrating remarkable structural stability. Nutrient release kinetics revealed sustained-release behavior, with cumulative elution of only 66.91% for urea and 92.45% for humic acid over 15 days. Under salt stress, the 1.5% CPAUH amendment (P2) markedly enhanced wheat growth compared with the non-amended control (P0), as reflected by significant increases in plant height, chlorophyll content, fresh weight, dry weight, and nitrogen uptake. Concurrently, CPAUH application effectively improved soil conditions by reducing electrical conductivity by 39.16% (to 4.38 mS·cm^{−1}). These collective findings of CPAUH hydrogel offer substantial potential as a multifunctional soil amendment for enhancing water-fertilizer efficiency, reclaiming saline–alkali soils, and improving crop productivity under resource-limited conditions.

Keywords: carboxylated cellulose nanofibers; hydrogel; slow release; water retention; soil amendment



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1. Introduction

With the rapid growth of the global population and societal progress, the availability of arable land is steadily declining, and water scarcity for agriculture has become increasingly severe in many regions. Consequently, there is an urgent need to enhance crop productivity to meet human demands [1–3]. Over the past few decades, agricultural intensification has relied heavily on fertilizers to sustain crop growth. However, excessive fertilizer application has caused serious environmental problems. In particular, a large fraction of nitrogen

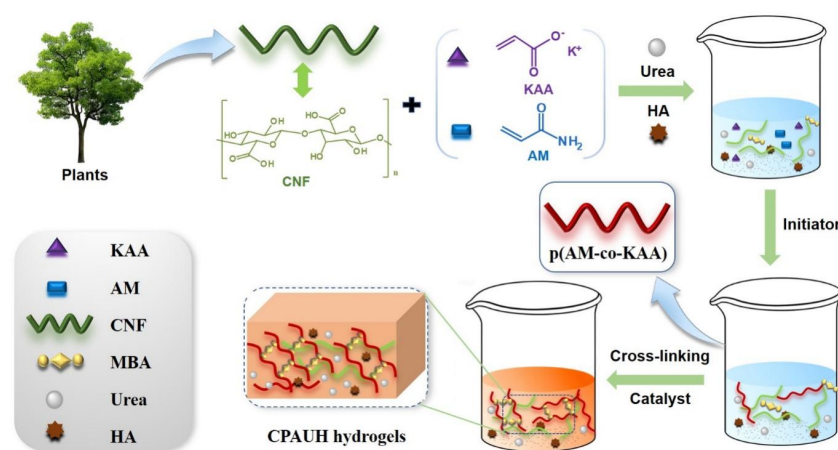
fertilizer that crops do not absorb volatilizes into the atmosphere as ammonia or nitrogen oxides, contributing to greenhouse effects; leaches downward as nitrate, contaminating groundwater; or is lost via surface runoff, ultimately causing water eutrophication. These pathways collectively constitute a cascade of environmental consequences resulting from nitrogen migration [4–6]. Therefore, it is necessary to develop efficient and environmentally friendly fertilizer technologies and carrier materials that enhance fertilizer use efficiency. Among these, slow-release fertilizers have shown clear advantages in reducing nitrogen loss, improving fertilizer utilization efficiency, and increasing crop yield [7–9]. Conventional slow-release technologies, such as coated fertilizers, are typically designed with a single primary function: modulating nutrient release rate. However, under complex field conditions such as salinity, alkalinity, and alternating dry-wet cycles, crop growth is simultaneously limited by multiple interacting factors. The unitary release-regulation approach is inadequate to address these intertwined constraints, which often restrict the efficacy and stability of traditional carriers in real-world agricultural environments. Consequently, the development of integrated smart carriers capable of synchronizing nutrient slow-release, water retention, and salt mitigation has emerged as a crucial technological pathway to enhance crop resilience and enable sustainable agriculture.

Hydrogels, as a class of polymer materials with a three-dimensional network structure, can absorb and retain substantial amounts of water. In recent years, hydrogel-based slow-release fertilizers have attracted increasing attention for their potential to improve nutrient use efficiency, conserve water, and promote crop growth [10–12]. From a soil improvement perspective, incorporating materials that enhance porosity or contain hydrophilic functional groups (e.g., hydroxyl or carboxyl groups) can improve soil water-holding capacity, enhance particle aggregation, and reduce water loss. Moreover, recent studies have emphasized the sustainable synthesis of composite hydrogels from agro-industrial wastes and natural polymers such as sodium alginate and bovine gelatin. Zhang et al. synthesized alginate-grafted polyacrylamide/Se–citrus peel polysaccharide hydrogels that improved selenium utilization efficiency compared to conventional Se fertilizers [13]. Khatun et al. prepared a polyvinyl alcohol/sodium alginate hydrogel cross-linked with citric acid, onto which urea was loaded. Humic acid addition further improved the hydrogel's sustained-release properties and promoted wheat growth [14]. Rebelo et al. developed biodegradable cellulose hydrogels from repurposed solvent salts as nitrogen fertilizers, which exhibited high swelling capacity, increased soil water retention, and delayed fertilizer release, aligning with sustainability goals for agricultural applications [15]. Lv et al. prepared biodegradable oxidized starch-based hydrogel as a slow-release carrier for urea, which exhibited a high water absorption capacity and sustained swelling after multiple cycles, significantly enhancing plant growth [16]. The current research has extensively explored water gels based on biobased polymers as fertilizer carriers, highlighting their biodegradability and sustained nutrient release capabilities. However, most studies have faced drawbacks, such as the complex synthesis of raw materials used to prepare the gels and the high energy consumption required for the heating synthesis process. Moreover, research on the regulatory effects of these hydrogels under salt-stress conditions is still limited, and maintaining their structural and functional stability under such conditions remains a challenge.

In the pursuit of high-performance hydrogel carriers, incorporating biomass wastes and nanomaterials has shown great promise. Among various nanofillers (e.g., nanoclay and graphene oxide), carboxylated cellulose nanofiber (CNF) was selected for its balanced advantages in functional compatibility, environmental sustainability, and structural reinforcement. CNF is sourced from renewable biomass and offers more defined biodegradability and lower environmental risk. Owing to their high aspect ratio (>100) and rigid network structure, CNF confers exceptional strength and toughness to composites, surpassing traditional biopolymers

such as carboxymethyl cellulose (CMC) or alginate [17–19]. The abundant surface carboxyl groups of CNF offer active sites for functionalization, enabling the introduction of nutrient-adsorbing or controlled-release moieties. Hydrogels incorporating CNF generally display outstanding water retention and swelling capacities, favoring nutrient and moisture conservation [20]. To fully exploit CNF's nanoreinforcement potential, it must be integrated into a stable and robust polymer network. A practical approach is via an interpenetrating polymer network (IPN), in which two or more cross-linked polymer networks interpenetrate and entangle at the molecular level without chemical bonding. This topology produces synergistic reinforcement through physical interlocking, significantly improving mechanical stability. For example, Zhu et al. synthesized starch-g-P(acrylic acid-co-acrylamide)/PVA semi-IPN hydrogels via aqueous polymerization, where PVA crystal formation enhanced mechanical strength through physical entanglement and hydrogen bonding [21]. Kim et al. reported that hybridizing alginate hydrogels with polyvinyl alcohol in a semi-IPN configuration preserved alginate's advantages while mitigating its drawbacks [22]. Likewise, Wang et al. successfully synthesized semi-IPN-based elastomers that exhibit a favorable balance between self-healing ability and mechanical strength due to dynamic hydrogen bonding [23]. The combination of CNF's nanoreinforcement and the stabilizing copolymer to form semi-IPN hydrogels is thus expected to yield multifunctional carriers capable of buffering salt stress and providing a sustained water and nutrient supply to plant roots.

This study employed a carboxylated cellulose nano-fiber/poly(acrylamide-co-potassium acrylate) semi-IPN hydrogel (CPAUH) as the core carrier. Under mild conditions at room temperature and atmospheric pressure, urea and humic acid were efficiently loaded via a one-pot synthesis [24], establishing a green, low-energy preparation strategy (Scheme 1). This reaction uses water as the solvent and proceeds at room temperature without external heating. Moreover, the entire process does not rely on heavy metal catalysts, thereby significantly reducing energy consumption and chemical pollution at the source. Based on this hydrogel material, the study aimed to address three key issues synergistically: (1) to achieve the controlled release of urea and humic acid to provide the foundation for precise improvement of nitrogen use efficiency and soil amendment performance; (2) to elucidate the hydrogel's water regulation and retention capabilities under water-saving and saline conditions to mitigate soil environmental stress effectively; and (3) to evaluate this material's performance in the actual agricultural environment, the study assessed the impact of the hydrogel on nutrient absorption efficiency, salt tolerance, and overall growth and development of wheat during germination and the seedling stage. These two growth stages of wheat are susceptible to salt stress. Thus, it provided theoretical and experimental support for the development of environmentally friendly materials for sustainable agriculture.



Scheme 1. Diagram of preparation for CPAUH.

2. Results and Discussion

2.1. Characterization

2.1.1. FTIR Spectroscopy and TG Analysis

Figure 1a shows the FTIR spectra of CPA, CPAU, and CPAUH hydrogels. The broad and intense absorption bands at approximately 3120 cm^{-1} and 3450 cm^{-1} correspond to the combined stretching vibrations of $-\text{OH}$ and $-\text{NH}-$ groups, confirming the presence of hydroxyl groups from CNF and amino groups ($-\text{NH}_2$) from acrylamide (AM). The further broadening of these peaks in CPAUH suggests the formation of intermolecular hydrogen bonds after the incorporation of urea and humic acid. The absorption band at 2932 cm^{-1} is attributed to $-\text{CH}_2-$ stretching vibrations in the polymer backbone of the CNF/p(AM-co-KAA) hydrogel. The band at 1672 cm^{-1} arises from the $\text{C}=\text{O}$ stretching vibration of carboxyl groups, while the band near 1614 cm^{-1} is associated with the $\text{p}-\pi$ conjugation between the nitrogen lone pair in the amide group and the carbonyl group. Additionally, the peaks at 1452 cm^{-1} and 1408 cm^{-1} correspond to the symmetric stretching vibration of carboxylate anions.

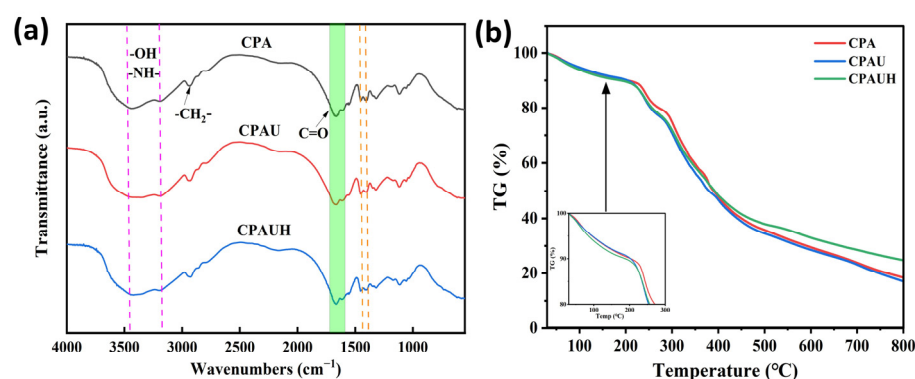


Figure 1. (a) FTIR spectra and (b) thermogravimetric analysis (TGA) curves of CPA, CPAU, and CPAUH hydrogels.

Figure 1b displays the TG curves of CPA, CPAU, and CPAUH hydrogels. The TGA profiles reveal distinct thermal decomposition behaviors for the three samples under a nitrogen atmosphere. An initial mass loss of approximately 10% occurs between $50\text{ }^{\circ}\text{C}$ and $200\text{ }^{\circ}\text{C}$, primarily due to the evaporation of residual moisture entrapped within the hydrogel network. A significant weight loss is observed in the $300\text{--}500\text{ }^{\circ}\text{C}$ range, corresponding to the decomposition of the polymer backbone. At $800\text{ }^{\circ}\text{C}$, the residual weights of CPA and CPAU are about 20% and 22%, respectively. The CPAU hydrogel shows slightly enhanced thermal stability, retaining approximately 30% of its weight at $800\text{ }^{\circ}\text{C}$, which suggests that the incorporation of urea does not substantially alter the thermodynamic behavior of the base CPA matrix. In contrast, CPAUH exhibits the highest thermal stability, attributed to the presence of humic acid, whose aromatic structures confer additional resistance to thermal degradation and enhance the overall thermal stability of the composite hydrogel.

2.1.2. Morphological Observation

The surface morphology of CPA, CPAU and CPAH gels after being soaked for 30 min was observed using a scanning electron microscope (SEM). As shown in the images, the structural characteristics of the hydrogels vary according to their composition. CPA and CPAU exhibit similar porous architectures, whereas CPAUH displays a distinct mesh-like structure. The surface pore sizes of the CPA hydrogel range from 6 to $12\text{ }\mu\text{m}$ (Figure 2a), while CPAU presents a denser and more compact pore structure with pores between 7 and $18\text{ }\mu\text{m}$ (Figure 2b). The CPAUH exhibits an interconnected mesh network

with pore sizes of approximately 7–14 μm (Figure 2c). The presence of these interconnected porous networks provides abundant internal space for urea and humic acid loading, thereby facilitating their diffusion and controlled release in aqueous and soil environments.

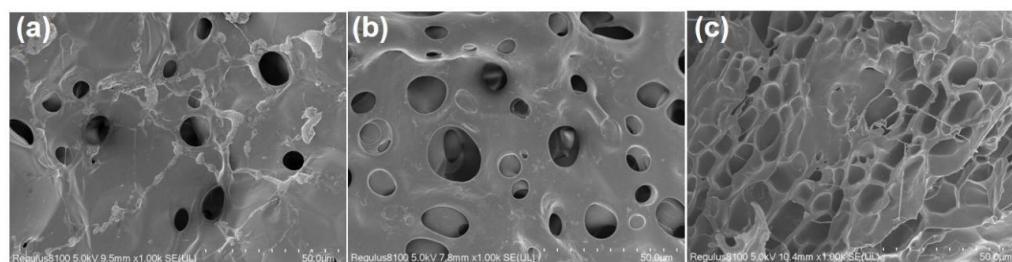


Figure 2. SEM images of (a) CPA, (b) CPAU, and (c) CPAUHs.

2.1.3. XPS Analysis

XPS was employed to determine the surface chemical composition of CPA and CPAUH hydrogels. The wide-scan spectra reveal three main peaks at 285 eV (C 1s), 400 eV (N 1s), and 532 eV (O 1s), corresponding to carbon, nitrogen, and oxygen, respectively (Figure 3a,d). Compared with CPA, the CPAUH hydrogel exhibits higher nitrogen (17.05%) and oxygen (24.33%) contents due to the successful incorporation of urea and humic acid. The high-resolution C 1s spectrum of CPAUH can be deconvoluted into four binding components: 284.80 eV (C–C/C–H), 287.87 eV (C–N/C–O), 292.59 eV (C=O), and 295.43 eV (O–C=O). Because the backbone structure of CPAUH is similar to that of CPA, their C 1s binding environments are largely consistent (Figure 3b,e). The O 1s spectrum confirms the coexistence of C=O (532.3 eV) and C–O (533.5 eV) functional groups in the hydrogel matrix. Notably, the C–O component area of CPAUH increases compared with CPA, attributable to the introduction of oxygen-rich humic acid (Figure 3c,f). These results collectively verify the successful incorporation of urea and humic acid within the CNF/p(AM-co-KAA) hydrogel network.

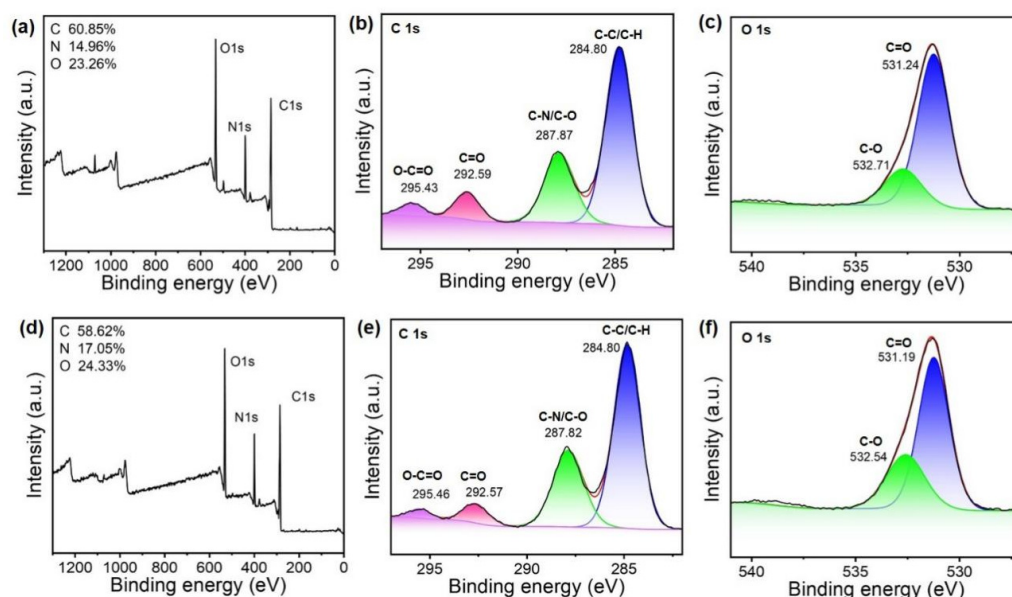


Figure 3. XPS wide-scan spectra, C 1s spectra, and O 1s spectra of CPA (a–c) and CPAUH (d–f).

2.1.4. Mechanical Properties of Hydrogels

The hydrophilicity and mechanical performance of the hydrogels were evaluated using dumbbell-shaped samples (Figure 4a). The contact angle of CPAUH was approximately 62° ,

higher than those of CCA, CPA, and CPAU, indicating relatively greater surface hydrophobicity. This effect likely arises from the aromatic rings present in humic acid molecules (Figure 4b). To systematically evaluate the influence of the semi-interpenetrating polymer network (semi-IPN) structure and nutrient loading on mechanical performance, tensile tests were conducted (Figure 4c,d). The incorporation of CNF significantly enhanced the tensile properties of the P(AM-co-KAA) hydrogel, increasing tensile stress from 105 kPa to 152 kPa and elongation at break from 295% to 384% (CCA vs. CPA). This improvement results from the introduction of CNF, which forms a semi-IPN structure through hydrogen bonding and physical entanglement, strengthening the polymer network beyond conventional chemical cross-linking. CNF can serve as both a gel scaffold and a reinforcing phase, forming a stable three-dimensional network through physical entanglement that significantly enhances mechanical properties. Upon urea loading, the tensile stress further increased from 152 kPa to 212 kPa, suggesting that hydrogen bonding between urea, CNF, and the P(AM-co-KAA) network contributes to a denser cross-linked structure. The surface carboxyl groups establish stable hydrogen bonds and electrostatic interactions, reinforcing the gel network and enhancing mechanical integrity. For CPAUH, which incorporates both urea and humic acid, a slight decrease in tensile strength was observed. The result is likely due to factors such as load quantity influencing gel strength. Nevertheless, the results confirm that the CNF-based semi-interpenetrating network substantially improves the mechanical properties of P(AM-co-KAA) hydrogels while maintaining flexibility and functional loading capacity.

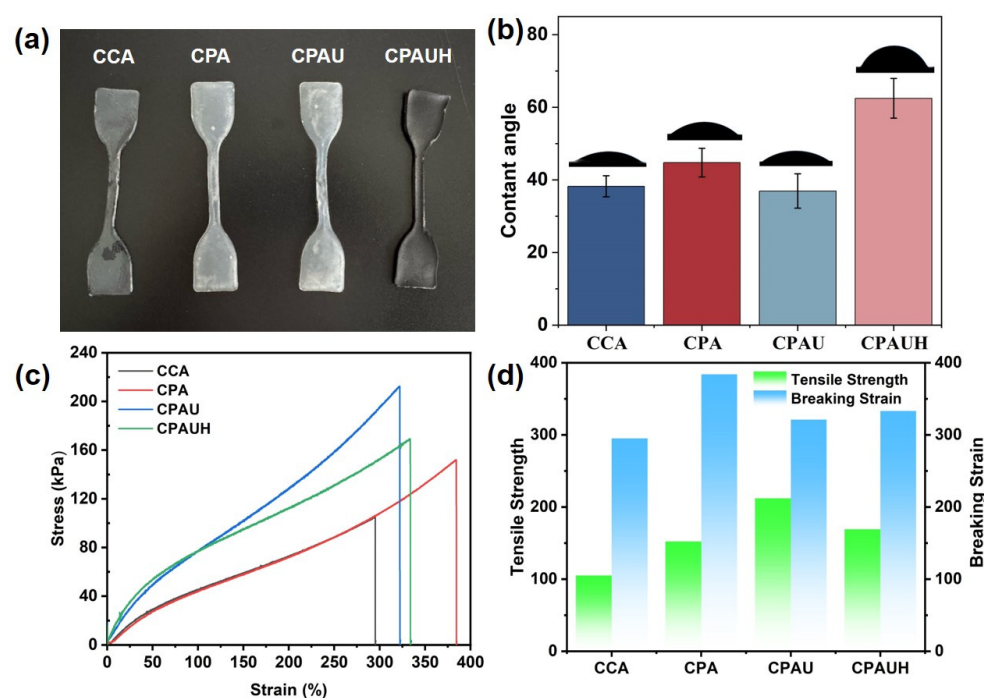


Figure 4. (a) Digital photos of the samples used for mechanical property testing, (b) water contact angle of the hydrogen samples, (c) tensile stress–strain curves, and (d) the tensile strength and breaking strain value of CCA, CPA, CPAU and CPAUH.

2.2. Performance

2.2.1. Swelling Capacity

The water absorption and swelling behavior of the hydrogels were investigated to evaluate their responsiveness to different environments [25]. As shown in Figure 5a, CPAUH reached a maximum swelling ratio of 121.65 g g^{−1} in distilled water after 24 h. During the initial 0–12 h period, the swelling ratio increased rapidly, after which the rate

gradually slowed and eventually stabilized. This kinetic behavior indicated a strong water absorption capacity in the early stage, followed by saturation as the network structure reached its expansion limit. The notably high initial swelling rate was likely attributable to the rapid formation of extensive hydrogen bonds between water molecules and the amide groups ($-\text{CONH}_2$) in the CPAUH molecular chains, coupled with the swift unfolding of polymer chains upon hydration, which created space for water penetration and storage [26]. After 12 h, the approach to equilibrium reflected a balance between the elastic retractive force of the fully swollen crosslinked network and the osmotic driving force, thereby restricting further expansion of the network.

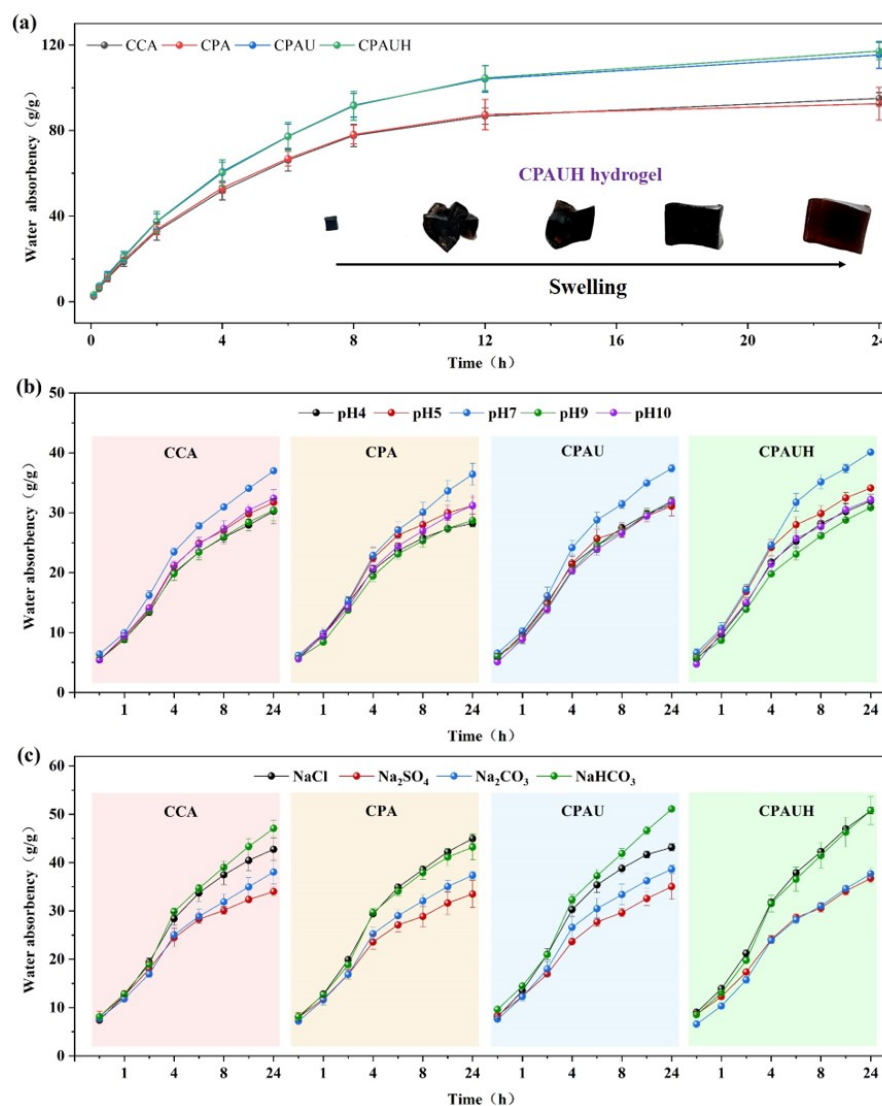


Figure 5. Swelling behavior of the CCA, CPA, CPAU, and CPAUH hydrogels: (a) Equilibrium swelling ratio (g g^{-1}) in distilled water, (b) Effect of pH on the swelling ratio, (c) Effect of different salt solutions on the swelling ratio.

Figure 5b compares the water absorption performance of the four materials under different pH conditions [27]. CPAUH exhibited optimal swelling behavior in a neutral environment (pH 7), with a swelling ratio of 40.10 g g^{-1} , while its swelling capacity decreased significantly under both acidic and alkaline conditions. This phenomenon demonstrated the distinct pH sensitivity of CPAUH. In an acidic medium, the abundant H^+ ions protonate the carboxylic acid functional group ($-\text{COO}^-$), converting it into a neutral carboxylic acid group ($-\text{COOH}$). This protonation removes the negative charge, thereby eliminating the

electrostatic repulsion between polymer chains. Additionally, hydrogen bonding between the $-\text{COOH}$ groups may further promote chain aggregation, leading to network contraction and reduced water absorption capacity. Under alkaline conditions, Na^+ ions shielded the negative charges on $-\text{COO}^-$ groups, weakening electrostatic repulsion between polymer chains, while the elevated ionic strength decreases the osmotic pressure difference [28]. These combined effects suppressed gel swelling. The distinct pH responsiveness of CPAUH thus indicated its tunable water absorption performance across different acid–base environments, supporting its potential application in soils or crop rhizospheres with specific pH conditions.

Figure 5c illustrates the swelling behavior of the four hydrogels in various sodium salt solutions (NaCl , Na_2SO_4 , Na_2CO_3 , NaHCO_3). All materials displayed significantly lower water absorbency in salt solutions than in distilled water, with markedly reduced swelling ratios in Na_2SO_4 and Na_2CO_3 compared to NaCl and NaHCO_3 . The overall decrease in swelling capacity under saline conditions was mainly attributed to the shielding effect of cations (e.g., Na^+) on the anionic carboxylate groups within the hydrogel network, which reduced electrostatic repulsion and restricted network expansion [29]. Notably, in salt solutions containing divalent anions (e.g., SO_4^{2-} , CO_3^{2-}), the higher Na^+ concentration per mole of salt intensified the charge-shielding effect, resulting in a greater reduction in water absorption than in monovalent salt systems. These results suggest that, in practical agricultural applications, the type and ionic strength of soil salts play a crucial role in determining the water-retention performance of hydrogels and serve as a key criterion for evaluating their suitability.

2.2.2. Water Absorption Kinetics

The resulting parameters for water absorption kinetics obtained using the two models are summarized in Table 1. The correlation coefficient R^2 for the pseudo-first-order model (R_1^2) was higher than that for the pseudo-second-order model (R_2^2), and the amount of water at equilibrium (Q_{1e}) obtained from the pseudo-first-order model was in better agreement with the experimental data (Q_{eq}), as compared to that (Q_{2e}) from the pseudo-second-order model. Thus, the pseudo-first-order model was used to evaluate the swelling behaviors of CCA, CPA, CPAU, and CPAUH in distilled water. Based on the pseudo-first-order kinetic analysis, sample CPAUH exhibits the smallest swelling rate constant (k) and the largest equilibrium swelling capacity (Q_e). This result indicates that while its water absorption rate is relatively slow, it achieves the highest water uptake capacity. For hydrogel fertilizers, slow-release properties and long-term water retention are critical: the slower swelling kinetics favor the gradual release of both water and nutrients, while the higher absorption capacity ensures superior water uptake and retention in soil.

Table 1. Swelling kinetic parameters of CCA, CPA, CPAU, and CPAUH.

	CCA	CPA	CPAU	CPAUH
Q_{eq}	94.99	92.54	115.31	117.10
Q_{1e} (g g^{-1})	95.32	93.31	116.21	118.14
K_1	0.20	0.22	0.19	0.18
R_1^2	0.999	0.999	0.999	0.999
Q_{2e} (g g^{-1})	114.94	109.89	140.85	142.86
K_2	0.0019	0.0023	0.0015	0.0014
R_2^2	0.995	0.994	0.993	0.991

2.2.3. Water Retention and Reswelling Performance

Figure 6a shows the water loss curve of CPAUH at 25 °C. The sample completed the dehydration process within 48 h, exhibiting a relatively constant dehydration rate throughout this period. This stable dehydration trend indicated that water evaporation was primarily governed by ambient conditions (e.g., temperature, humidity) rather than by the material's intrinsic diffusion resistance. This behavior suggested a relatively uniform pore size distribution within the hydrogel network, allowing water to diffuse out at a consistent rate [30,31]. Such controllable dehydration was significant for practical agricultural applications, as it could facilitate a steady release of water in alternating dry-wet environments, thereby providing a more sustained water supply for crops.

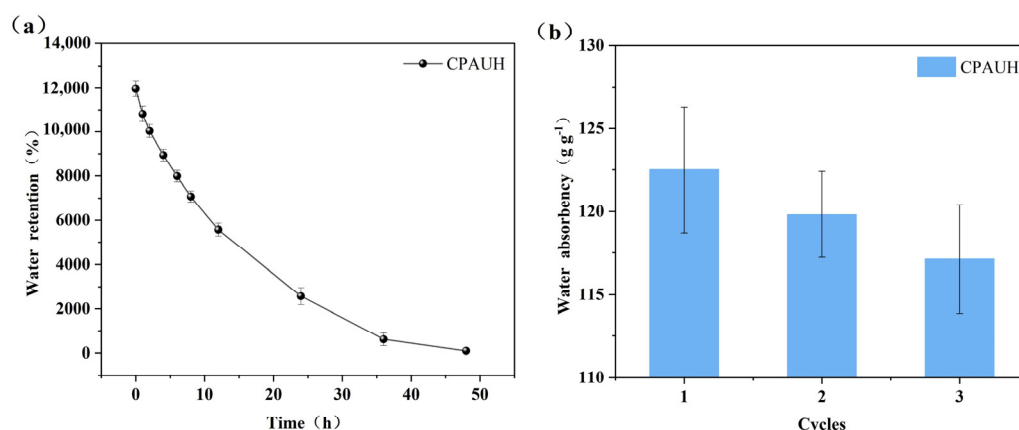


Figure 6. (a) Water retention curve of the CPAUH, (b) Reswelling capacity of the CPAUH hydrogel over three cycles.

The reusability of CPAUH was evaluated over three successive desorption-absorption cycles, as presented in Figure 6b. The swelling ratio displayed a gradual decrease with increasing cycle number but remained at a high level of approximately 118 g g⁻¹ even after the third cycle. The gradual reduction in swelling ratio was primarily attributed to mechanical fatigue within the polymer network induced by repeated swelling and contraction [32]. Each cycle involved the extension and retraction of crosslinked chains, potentially leading to the irreversible disruption of weaker physical crosslinks (e.g., hydrogen bonds, chain entanglements) or the cleavage of chemical crosslinks, resulting in slight network degradation and a permanent loss of water absorption capacity. Despite this, CPAUH retained about 97% of its initial swelling ratio (121.65 g g⁻¹) after three cycles, demonstrating excellent mechanical resilience and structural stability of its crosslinked network. This result confirms the outstanding reliability and long-term potential of CPAUH as a reusable material for water management.

2.2.4. Urea and Humic Acid Release Performance

To evaluate the environmental application potential of the prepared gel fertilizer, the nutrient release kinetics were systematically investigated under hydroponic conditions at 25 °C. The result shows the cumulative release profiles of urea and humic acid over 15 days. Both nutrients exhibited a typical biphasic release pattern. Within the initial 10 days, approximately 66.91% of urea and 92.45% of humic acid were released (Figure 7a,b). The difference in cumulative release rates between urea and humic acid is due to the fundamentally different existence states and release mechanisms in the gel matrix. The release of urea is mainly driven by physical diffusion but is restricted by interactions within the gel network, leading to incomplete release. Humic acid, as a highly water-soluble

substance, is more likely to achieve complete release. They can work together to exert their functions of improving the rhizosphere environment and promoting crop growth.

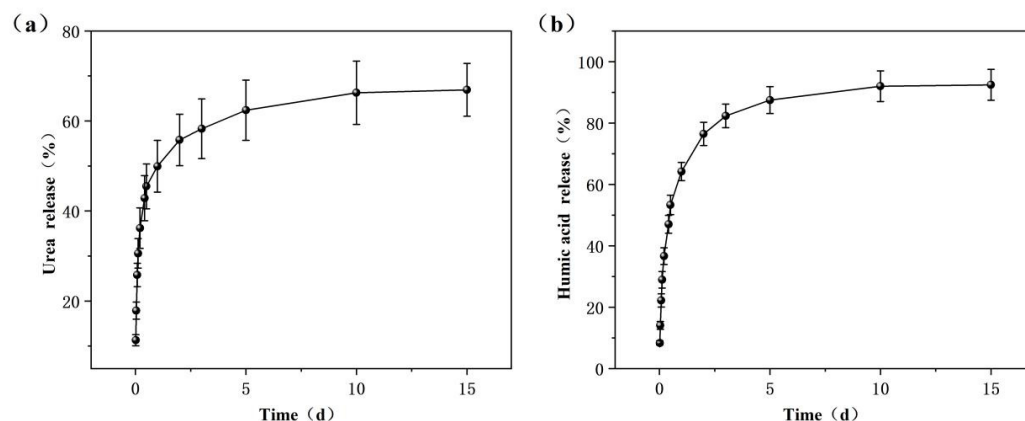


Figure 7. Cumulative release profiles of (a) urea and (b) humic acid from the CPAUH.

The initial rapid release was attributed to the readily soluble nutrients that were physically adsorbed on the gel surface or in macropores, which dissolved quickly and diffused out via short paths upon contact with water. This feature was agronomically advantageous, as it could provide immediate and essential starter nutrients for seed germination and early seedling growth, ensuring nutritional support during the critical initial stages [33]. The significant slowdown and eventual stabilization of the release rate after 10 days were mainly due to two factors: First, the rapid depletion of nutrients from the gel surface and easily accessible regions led to longer diffusion paths and increased resistance for internal nutrients, reducing the concentration gradient driving diffusion; Second, the release mechanism gradually shifted from initial dissolution-dominated release to a sustained-release mode controlled by the swelling degree and degradation behavior of the gel network. Over time, the gel carrier might undergo slight biodegradation or erosion, forming new diffusion pathways that collectively regulate the slow nutrient release in the later stage [34]. This biphasic release pattern, rapid initially and sustained afterward, aligned well with the nutrient demand pattern of crops, which was high during early growth stages and more stable later on. It demonstrated that the CPAUH gel fertilizer held strong potential for controlling nutrient release rates and improving fertilizer use efficiency.

2.2.5. Improvement Effect of Hydrogel on Wheat Growth

To investigate the salt resistance, water retention, and slow-release fertilizer capabilities of CPAUH, different fertilization treatments were established to elucidate its growth-promoting effect on wheat and its underlying mechanism in improving soil physicochemical properties.

Plant height data demonstrated CPAUH-mediated alleviation of salt stress (Figures 8a and 9). At 21 days, high-dose CPAUH (P2) promoted growth to 12.5 cm, substantially exceeding the control (CK, 7.9 cm), indicating early physiological regulation. By day 35, P2 exhibited overcompensation growth (22.8 cm), significantly surpassing CK (19.3 cm) and the strongly inhibited P0 (11.8 cm). This established a consistent growth gradient (P2 > CK > P1 > P0), confirming a dose-dependent synergistic effect. Mechanistically, CPAUH acted through three pathways: physically improving soil structure and water retention [35,36]; chemically adsorbing Na^+ via ion exchange to mitigate ionic toxicity [37]; and physiologically enabling sustained nutrient release [38]. The overcompensation in P2 likely stemmed from this combined stress mitigation and rhizosphere optimization, which activated compensatory

growth through resource reallocation and hormonal regulation [39,40]. Therefore, effective alleviation of salt stress by CPAUH requires sufficient dosage.

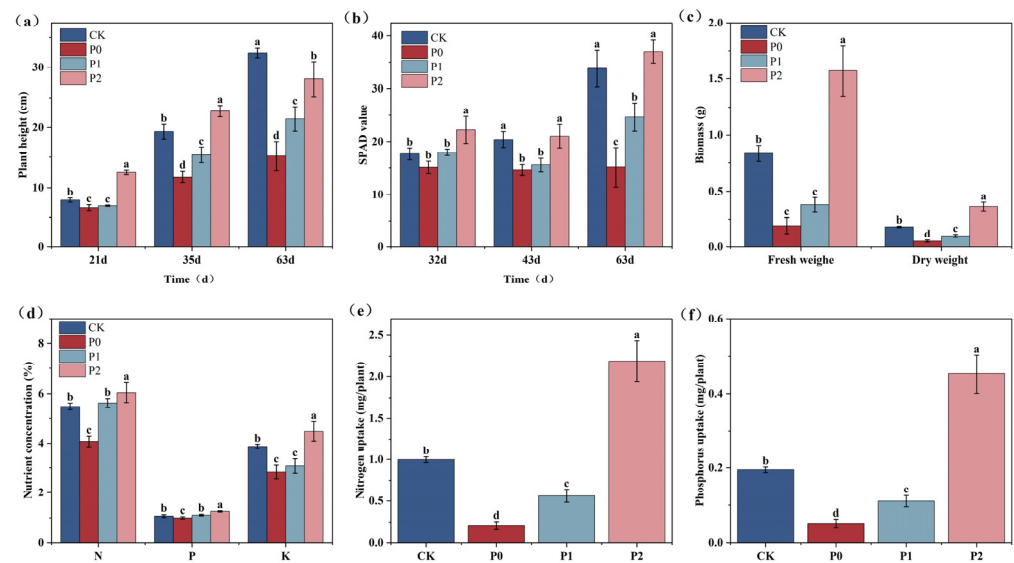


Figure 8. Plant growth and nutrient uptake parameters: (a) plant height, (b) chlorophyll content (SPAD value), (c) fresh and dry weight, (d) total nitrogen, phosphorus, and potassium content, (e) nitrogen uptake, and (f) phosphorus uptake. Bars ($n = 3$) with different letters indicate significantly different according to one-way ANOVA followed by Tukey's HSD test at $p < 0.05$.

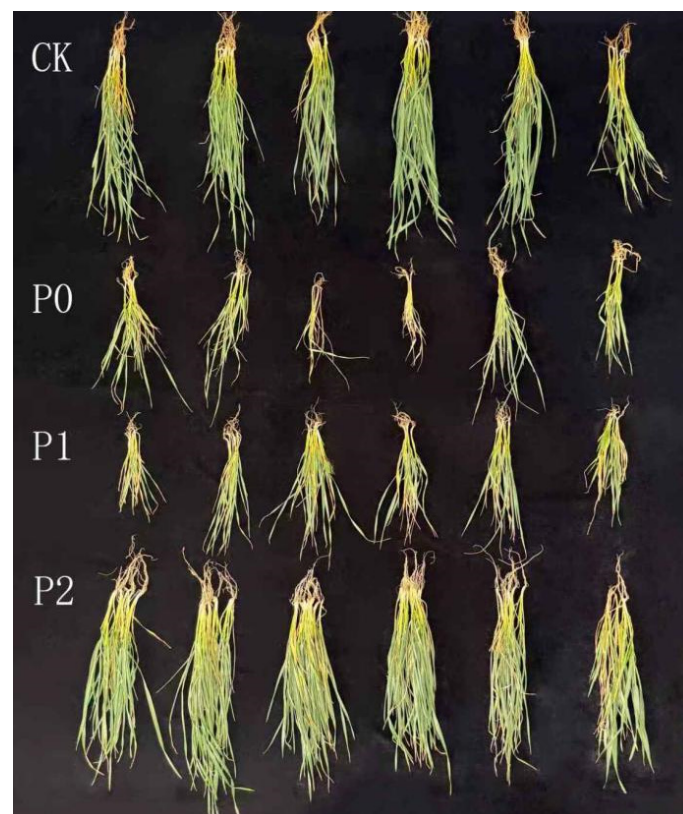


Figure 9. Digital photos of wheat plant phenotypes.

Chlorophyll content dynamics reflected both salt stress damage and CPAUH restoration (Figures 8b and 9). At 32 days, chlorophyll in P0 (15.09) was significantly lower than in CK (17.64), indicating substantial chloroplast damage, whereas P2 (22.23) demonstrated

CPAUH's protective role. This protection operated through three levels: (1) cellular, by maintaining ionic homeostasis to alleviate membrane lipid peroxidation; (2) metabolic, via sustained nutrient release for chlorophyll biosynthesis; and (3) systemic, likely by enhancing reactive oxygen species (ROS) scavenging through modulated antioxidant enzymes [41]. The dynamics further revealed an active adaptation strategy. By 43 days, CPAUH-treated plants transitioned from passive damage mitigation to active adaptation. At 63 days, P2 (37.03) showed no significant difference from CK (33.81), indicating that CPAUH not only protected against salt damage but also stimulated the plant's intrinsic self-repair capacity. Maintaining higher chlorophyll content enhanced light capture efficiency, which may explain the greater plant height in P2 [42]. The consistently elevated chlorophyll in P2 across all stages aligned with its superior growth, reinforcing that photosynthetic performance determines growth outcomes. This provides physiological evidence that CPAUH, at sufficient dosage, both mitigates stress and activates endogenous adaptation in saline environments.

Biomass data confirmed the alleviating effect of CPAUH under salt stress (Figures 8c and 9). Salt stress severely inhibited growth: the fresh and dry weights of P0 (0.19 g and 0.05 g) were only 22.9% and 27.8% of CK (0.83 g and 0.18 g), respectively, indicating suppressed water uptake and photoassimilate accumulation. CPAUH application reversed this inhibition. The P2 treatment achieved fresh and dry weights of 1.57 g and 0.36 g, reaching 189.2% and 200.0% of CK and significantly exceeding the control. The higher fresh weight in P2 suggests CPAUH mitigated osmotic stress and enhanced water uptake and retention, potentially via improved root development and soil structure [43]. This improvement in water relations aligns with the sustained increase in plant height, confirming CPAUH's role in maintaining water balance. Regarding dry matter, the twofold greater dry weight in P2 versus CK indicates that CPAUH alleviated photosynthetic inhibition and promoted carbon assimilation and partitioning. This enhancement likely resulted from (1) a stable water supply and reduced ion toxicity sustaining photosynthesis; (2) continuous nutrient release supporting photosynthetic function; and (3) reduced energy cost for osmotic adjustment, redirecting photoassimilates to growth [44]. Notably, P2 surpassed CK in both biomass parameters, suggesting that optimal soil amendment can not only counteract stress but also stimulate growth potential [45], possibly through CPAUH-improved rhizosphere conditions that regulate endogenous hormones and optimize photoassimilate use. In summary, this study confirms that high-dose CPAUH effectively promotes biomass accumulation in saline soil, providing a theoretical and practical basis for its use in remediation.

Total nitrogen (N), phosphorus (P), and potassium (K) content and uptake dynamics demonstrated CPAUH's capacity to enhance nutrient acquisition under salt stress (Figure 8d–f). Salt stress reduced nutrient uptake, as seen in the lower total N in P0 (4.08%) versus CK (5.49%), likely due to Na^+ -induced inhibition of transporters and reductase activity. In contrast, P2 not only alleviated this constraint but exceeded CK in total N (6.04%) and N uptake, indicating pronounced nutrient promotion. This enhancement reflects CPAUH's multi-regulatory role. In N metabolism, “nitrogen superaccumulation” in P2 likely resulted from (1) Na^+ adsorption alleviating ionic toxicity and improving transporter activity; (2) sustained N release ensuring supply; and (3) rhizosphere optimization promoting root development and absorption area [46]. For P, CPAUH improved both content and uptake, possibly by modifying the rhizosphere and mobilizing soil-fixed P [47]. K dynamics were particularly notable: P2's performance confirmed CPAUH's effectiveness in mitigating Na^+ -induced competitive inhibition of K^+ uptake, thereby maintaining ion balance and osmotic regulation [48]. Nutritionally, the synergistic promotion of N, P, and K uptake in P2 helped re-establish metabolic balance, consistent with the growth advantages in height and biomass, underscoring CPAUH's role in sustaining nutrient homeostasis under salinity.

Soil electrical conductivity (EC), a key indicator of soil salinity, revealed CPAUH's significant ameliorative effect (Figure 10a). Compared to the control (CK), the salt-stressed treatment (P0) exhibited a markedly higher EC, exceeding CK by 3.02 mS cm^{-1} . CPAUH application showed a dose-dependent response: while P1 was similar to P0, the P2 treatment achieved a substantially lower EC, representing a reduction of 4.38 mS cm^{-1} versus P0. This pronounced reduction in P2 stemmed from its multi-scale remediation. The hydrogel's network structure improved soil porosity to facilitate salt leaching, while its functional groups directly reduced soil solution ion concentration via ion exchange. This physicochemical synergy effectively improved the rhizosphere microenvironment [49]. The regulated EC in P2 corresponded closely with its superior plant height and biomass, confirming that rhizosphere modification through material science is a viable strategy for salt stress mitigation. This synergistic “material-structure-function” mechanism offers considerable promise for sustainable soil management.

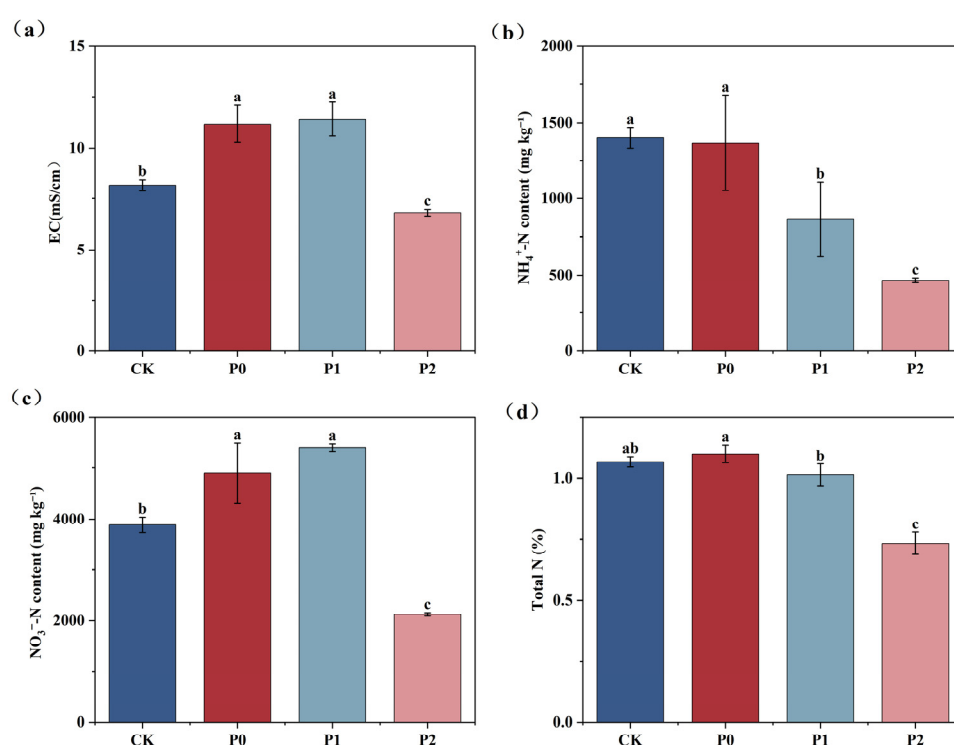


Figure 10. Soil physicochemical properties: (a) electrical conductivity (EC), (b) ammonium nitrogen ($\text{NH}_4^+\text{-N}$), (c) nitrate nitrogen ($\text{NO}_3^-\text{-N}$), (d) total nitrogen. Bars ($n = 3$) with different letters indicate significantly different according to one-way ANOVA followed by Tukey's HSD test at $p < 0.05$.

Figure 10b–d illustrate how different treatments regulated soil nitrogen forms, highlighting CPAUH's role in modulating the nitrogen cycle. Ammonium nitrogen ($\text{NH}_4^+\text{-N}$) content in the salt-stressed treatment (P0) did not differ significantly from CK, whereas CPAUH application reduced $\text{NH}_4^+\text{-N}$ in a dose-dependent manner (P1 and P2 decreased by 535.99 and 935.68 mg kg^{-1} relative to CK). In contrast, nitrate nitrogen ($\text{NO}_3^-\text{-N}$) was significantly higher in P0 and P1 than in CK (increases of 1011.33 and 1512.35 mg kg^{-1} , respectively), but was significantly lower in P2. Total nitrogen was also lowest in P2 (0.73%), forming a clear “low-residual” nitrogen pattern, indicative of systematic regulation of nitrogen transformation. This synchronized reduction in all nitrogen forms under P2 reflects an optimized nitrogen flux in the soil–plant system. The effect likely stems from CPAUH-mediated improvements in the rhizosphere, which enhanced both nitrification and plant nitrogen uptake, thereby promoting efficient transfer from soil to plant [50]. This “promoted transformation-enhanced uptake” synergy shortened soil nitrogen residence

time and reduced potential loss. From an agroecological perspective, the nitrogen allocation pattern under P2 is valuable: low soil nitrogen residue alongside meeting crop demand suggests improved fertilizer use efficiency and lowered risk of non-point pollution [51,52]. Thus, from a nitrogen-cycling standpoint, high-dose CPAUH effectively optimizes nutrient management under salt stress. Future studies could use ^{15}N tracing to precisely quantify transformation pathways and allocation under CPAUH, supporting the development of precise fertilization strategies.

3. Conclusions

This study developed a multifunctional composite hydrogel synthesized from biomass through a facile one-pot method in water. The material exhibited remarkable potential for agricultural applications by simultaneously enhancing water conservation, nitrogen retention, and soil amelioration. CPAUH semi-IPN hydrogel was constructed and incorporated functional components (urea and humic acid), endowing the hydrogel with a well-developed porous structure, tunable wettability, favorable mechanical strength, and abundant nitrogen- and carbon-containing groups. The synergistic interplay of these physicochemical properties resulted in outstanding performance, including high water retention and reswelling capacity, controlled nutrient release, resistance to salt stress, promotion of plant growth, and improvement of soil fertility—demonstrating its suitability for practical agricultural use. Moreover, the straightforward synthesis under ambient conditions further underscored its scalability for industrial production. This work highlighted the critical role of the CPAUH as a fertilizer carrier, enabling synergistic slow-release, water-retentive, and salt-resistant properties. Our findings provide a robust framework for developing advanced agricultural amendments and fertilizer carriers, offering a viable strategy to mitigate drought and abiotic stress while promoting efficient and sustainable agricultural practices.

4. Materials and Methods

4.1. Materials

Acrylamide (AM, 99%), potassium acrylate (KAA, 95%), urea, humic acid ($\text{HA} \geq 90\%$), carboxylated cellulose nanofiber (CNF, diameter: 4–10 nm, length: ~200 nm), N, N'-methylenebisacrylamide (MBA), N, N, N', N'-Tetramethylethylenediamine (TMED) and ammonium persulfate (APS) were purchased from Macklin Biochemical Technology Co., Ltd. (Shanghai, China). Sodium chloride, sodium sulfate, sodium carbonate, and sodium bicarbonate were obtained from Sinopharm Chemical Reagent Co., Ltd. (Beijing, China). All reagents were of analytical grade. Phosphate-buffered saline (PBS) was purchased from Pinggen Technology Co., Ltd. (Yichun, China).

4.2. Synthesis of the Hydrogels

72 g of AM and 14.4 g of KAA were dissolved in the deionized water. After that, 0.48 g of MBA was added as the crosslinker, and 1.6 g of APS was added as the initiator to the solution. After thorough mixing, 200 μL of catalyst TMED is slowly added, and the hydrogel is initiated at room temperature to form the CCA hydrogel. After 8 g of CNF was added to prepare the CNF/p(AM-co-KAA), denoted CPA hydrogel. During the gel preparation process, 8 g of urea was incorporated into the CPA to form the urea-loaded CPAU hydrogel. Co-loading both 8 g of urea and 8 g of humic acid produced the final CNF/p(AM-co-KAA)/Urea/Humic acid composite hydrogel, named as CPAUH.

4.3. Swelling Studies

The swelling ratio (SR) of the hydrogels was determined gravimetrically. Dried hydrogels (W_d) were prepared by oven-drying at 60 °C to constant weight and then immersed in distilled water, PBS buffers (0.01 mol L⁻¹, pH 3, 5, 7, 9, 10), or salt solutions (0.01 mol L⁻¹ NaCl, Na₂SO₄, Na₂CO₃, NaHCO₃). Samples were removed at intervals (0.5, 1, 2, 4, 8, 12, and 24 h), blotted to remove surface water, and weighed (W_s). Each test was performed in triplicate. The equilibrium swelling ratio was calculated using Equation (1):

$$Q_{eq} = (W_s - W_d)/W_d \quad (1)$$

4.4. Swelling Kinetics

The CCA, CPA, CPAU, and CPAUH were immersed in an excess of distilled water at ambient temperature at different time (t) intervals (0.0883, 0.25, 0.5, 1, 2, 4, 6, 8, 12, 24 h). Then the Q_{eq} values at different times were fitted using the pseudo-first-order (Equation (2)) and the pseudo-second-order (Equation (3)) model, respectively:

$$Q_t = Q_{e1} (1 - e^{-k_1 t}) \quad (2)$$

$$t/Q_t = t/Q_{e2} + 1/(k_2 Q_{e2}^2) \quad (3)$$

where Q_e and Q_t (g g⁻¹) are the amounts of water adsorbed at equilibrium swelling and at time t , respectively; k_1 and k_2 are the rate constants of pseudo-first-order and pseudo-second-order kinetic equations, respectively.

4.5. Water Retention and Reswelling Capacity

To assess water retention, fully swollen hydrogels were air-dried at room temperature, and their weight was recorded periodically. The water retention ratio (R_i , %) was calculated using Equation (4):

$$R_i = (M_i/M_0) \times 100, \quad (4)$$

where M_0 and M_i represent the initial and time-dependent weights, respectively.

For reswelling analysis, dried hydrogels were re-immersed in distilled water until equilibrium was reached, then dried again to constant weight. This swelling–drying cycle was repeated three times, and the swelling ratio for each cycle was determined as above to evaluate structural resilience.

4.6. Nutrient Release Behavior

Nutrient release kinetics were studied by incubating 0.5 g of dry hydrogel in 100 mL of distilled water at 25 °C. At scheduled intervals, the entire solution was replaced with fresh distilled water. The withdrawn aliquots were diluted to 100 mL for quantitative analysis. The urea concentration was determined spectrophotometrically using dimethylaminobenzaldehyde at 430 nm by spectrophotometer. The concentration of humic acid was determined by a UV-Vis spectrophotometer at 465 nm.

4.7. Pot Experiment

The pot experiment was conducted in a wheat greenhouse at the Beijing Academy of Agriculture and Forestry Sciences. The soil was placed in pots with dimensions of 52 × 35 × 15 cm. The initial physicochemical properties of the soil were as follows: pH 5.76, electrical conductivity (EC) 5.68 mS cm⁻¹, total nitrogen 1.70%, available phosphorus 561.79 mg kg⁻¹, and available potassium 1076.00 mg kg⁻¹. Four treatments (CK, P0, P1, P2) in triplicate. Each pot contained 10 kg of air-dried soil. The CPAUH was applied at 0% (CK, P0), 0.5% (P1), and 1.5% (P2) by soil weight. Total nitrogen was equalized

across treatments with supplementary urea. A 2000 mL aliquot of 200 mM saline solution was added to P0, P1, and P2; CK remained untreated. The resulting soil ECs one day post-salinization were 11.32, 12.13, and 12.06 mS cm⁻¹ for P0, P1, and P2, respectively. During wheat growth, P1 and P2 received only 70% of the irrigation water applied to CK and P0.

The saline solution (200 mM) was prepared by dissolving 11.688 g of NaCl, 28.408 g of Na₂SO₄, 21.200 g of Na₂CO₃, and 16.802 g of NaHCO₃ in distilled water. The solution was then transferred into a 1000 mL volumetric flask and made up to volume with distilled water. The pH and electrical conductivity (EC) of the solution were 9.63 and 107.17 mS cm⁻¹, respectively.

4.8. Characterizations

Before freeze-drying, freshly synthesized hydrogels were immersed in deionized water for 1 h. The microstructure of the lyophilized samples was examined by scanning electron microscopy (SEM, Hitachi S4800, Hitachi High-Tech Corporation, Tokyo, Japan). Fourier transform infrared (FTIR, Thermo Scientific Nicolet iS20, Thermo Fisher Scientific Inc., Waltham, MA, USA) spectra were recorded in the range of 400–4000 cm⁻¹. The surface elemental composition was analyzed by X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha, Thermo Fisher Scientific Inc., Waltham, MA, USA), and the thermal stability was evaluated using thermogravimetric analysis (TG, NETZSCH STA449C, NETZSCH-Gerätebau GmbH, Selb, Germany).

At the wheat seedling stage, plant height was measured, and leaf chlorophyll content was assessed using a SPAD meter (TYS-A, Zhejiang Top Cloud-agri Technology Co., Ltd., Hangzhou, China). Soil volumetric water content was determined with a TDR350 probe (Model TDR350, Spectrum Technologies, Inc., Plainfield, IL, USA). At harvest, fresh and dry biomass were recorded. Plant tissues were digested and analyzed for nutrients: total N (Kjeldahl method), total P (UV spectrophotometry), and total K (flame photometry). Post-harvest soil samples were air-dried and sieved. The following were performed: total N (Kjeldahl digestion with H₂SO₄), available P (NaHCO₃ extraction–molybdenum–antimony colorimetry), available K (NH₄OAc extraction–flame photometry), soil organic matter (K₂Cr₂O₇ titration), pH (2.5:1 water-soil), and EC (5:1 water-soil).

4.9. Data Analysis

Data were processed using Excel 2019, SPSS 22.0, and Origin 2024 software. Differences among treatments were evaluated by one-way ANOVA followed by multiple comparison tests, with statistical significance set at $p < 0.05$.

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